Left-closed Sets

§ **Left-closed**

Parameters

+ $A \in \text{Set}$
+ $R \in A \sim A$
+ $X \in \text{Set}$

Assumptions

a. $X \subseteq A$

b. $(\forall a, b \mid a \mathrel{\mathrm{R}} b \wedge b \in X : a \in X)$

Observations

1. $X = R[X] \Rightarrow \text{Left-closed } R\ X$
2. $X = R[X] \Leftarrow \text{Left-closed } R\ X \wedge \text{Reflexive } R$
3. $(\text{Left-closed } R\ X \equiv X = R[X]) \Leftarrow \text{Reflexive } R$

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Proof of Observation 1.

◆ $X = R[X]$

$\equiv$ Extensionality

$(\forall a \mid : a \in X \equiv a \in R[X])$

$\Rightarrow$ Predicate calculus

$(\forall a \mid a \in R[X] : a \in X)$

$\equiv$ Direct image

$(\forall a \mid (\exists x \mid x \in X : a \mathrel{\mathrm{R}} x) : a \in X)$

$\equiv$ Splitting

$(\forall x \mid x \in X : (\forall a \mid a \mathrel{\mathrm{R}} x : a \in X))$

$\equiv$ Nesting

$(\forall a, x \mid x \in X \wedge a \mathrel{\mathrm{R}} x : a \in X)$

$\equiv$ Definition of Left closed

$\text{Left closed } R\ X$

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Proof of Observation 2.

◆ $X = R[X]$

$\equiv$ Extensionality

$(\forall a \mid : a \in X \equiv a \in R[X])$

$\equiv$ Mutual implication

$(\forall a \mid a \in X : a \in R[X]) \wedge$

$(\forall a \mid a \in R[X] : a \in X)$

$\equiv$ Definition of Left closed

$(\forall a \mid a \in X : a \in R[X]) \wedge$

Left set R X

$\equiv$ Identity of $\wedge$

◆ Left closed R X

… $(\forall a \mid a \in X : a \in R[X])$

$\equiv$ Definition of direct image

$(\forall a \mid a \in X : (\exists x \mid x \in X : a \mathrel{\mathrm{R}} x))$

$\equiv$ Reflexivity of R

◆ Reflexive R

… true

■

Proof of Observation 3.

◆ $X = R[X]$

i. Reflexive R

$\Rightarrow$ Observation 1.

Left-closed R X

$\equiv$ Assumption i.

Left-closed R $X \wedge$ Reflexive R

$\Rightarrow$ Observation 2.

$X = R[X]$

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